

§ 2.5 Convergence of random series

Def 1 tail σ -field

Let $\mathcal{F}_n' = \sigma(X_n, X_{n+1}, \dots)$ the smallest σ -field $X_m (m \geq n)$ are measurable

$\mathcal{T} = \bigcap_n \mathcal{F}_n'$ is tail σ -field.

$A \in \mathcal{T} \Leftrightarrow \forall n$ the occurrence of A is only determined by $(X_n, X_{n+1}, \dots)$
not affected by $(X_1, \dots, X_{n-1})$

Borel-Cantelli Lemma I
 $\sum_n P(A_n) < \infty$ then $P(A_n \text{ i.o.}) = 0$

B-C Lemma II
 $A_1, A_2, \dots$ i.i.d. $\sum_{n=1}^{\infty} P(A_n) = \infty$ then
 $P(A_n \text{ i.o.}) = 1$

Example 1: $S_n = X_1 + X_2 + \dots + X_n$, then $\{\lim_{n \rightarrow \infty} S_n \text{ exists}\} \in \mathcal{T}$

THM 1 (Kolmogorov's 0-1 law)

If $X_1, X_2, \dots$ are independent and $A \in \mathcal{T}$ then $P(A) = 0$ or 1

proof: Consider that $P(A|A) = P(A)$ A is independent with itself.

Let $\mathcal{F}_n = \sigma(X_1, X_2, \dots, X_n)$ $\mathcal{F}_n' = \sigma(X_{n+1}, X_{n+2}, \dots)$ $\mathcal{T} = \bigcap_n \mathcal{F}_n'$

$\{X_n\}$ i.i.d. $\Rightarrow \forall n$ $\mathcal{F}_n, \mathcal{F}_n'$ i.i.d. $\because A \in \mathcal{T} \subset \mathcal{F}_n' \therefore A \in \mathcal{F}_n'$

$\Rightarrow A, \mathcal{F}_n$ i.i.d. Let $\mathcal{F}_{\infty} = \sigma(\bigcup_{n=1}^{\infty} \mathcal{F}_n) \Rightarrow A, \mathcal{F}_{\infty}$ i.i.d.

and $A \in \mathcal{T} \subset \mathcal{F}_{\infty} \Rightarrow A, A$ i.i.d. $\Rightarrow P(A) = 0$ or 1 . (Q)

Def 2 Finite permutation and permutable event

If $\pi: \mathbb{N} \rightarrow \mathbb{N}$ $\pi(i) \neq i$ for only finitely many i , then we call it finite permutation.

If $\pi^{-1}(A) \equiv \{\omega: \pi\omega \in A\}$ is equal to A for any permutation π we can the event A is permutable.

Remark: ① The collection of permutable events is a σ -field and it is called exchangeable σ -field and denoted by Σ

② All events in the tail σ -field $\mathcal{T}$ are permutable, $\mathcal{T} \subset \Sigma$

Thm 2 LH-S 0-1 law

If $X_1, \dots, X_n$ are i.i.d and $A \in \Sigma$ then $P(A) \in \{0, 1\}$

Proof the idea is as same as K 0-1 law

Let $A_n \in \sigma(X_1, \dots, X_n)$ s.t. $P(A_n \Delta A) \xrightarrow{\text{对称}} 0$ $\pi(j) = \begin{cases} j+n & 1 \leq j \leq n \\ j-n & n+1 \leq j \leq 2n \\ j & j \geq 2n+1 \end{cases}$

the coordinates and permuted coordinates i.i.d

$$P(\omega: \omega \in A_n \Delta A) = P(\omega: \pi\omega \in A_n \Delta A) = P(\omega: \omega \in A_n' \Delta A)$$

$$P(A_n \Delta A_n') \leq P(A_n \Delta A) + P(A_n' \Delta A) \rightarrow 0 \text{ that means when } n \text{ is}$$

$$\text{large, } A_n \text{ is as same as } A_n' \Rightarrow P(A_n \cap A_n') \rightarrow P(A)$$

$$\text{From } A_n, A_n' \text{ i.i.d.} \Rightarrow P(A_n \cap A_n') = P(A_n)P(A_n') \rightarrow P(A)^2 = P(A) \quad \square$$

Lemma 2 (Kolmogorov's maximal inequality)

Suppose $X_1, \dots, X_n$ i.i.d. with $EX_i = 0$ and $\text{Var}(X_i) < \infty$ $S_n = X_1 + \dots + X_n$

then $P(\max_{1 \leq k \leq n} |S_k| \geq x) \leq x^{-2} \text{Var}(S_n)$

Compared with Chebyshev inequality that $P(|S_n| \geq x) \leq x^{-2} \text{Var}(S_n)$

the proof is ignored

THM 3

Suppose that $X_1, X_2, \dots$ i.i.d. $EX_n = 0$ If $\sum_{n=1}^{\infty} \text{Var}(X_n) < \infty$
then $\sum_{n=1}^{\infty} X_n(\omega)$ converges with probability 1.

proof. Let $S_N = \sum_{n=1}^N X_n$ from Lemma 1 we have

$$P(\max_{m \leq n \leq N} |S_m - S_n| > \varepsilon) \leq \varepsilon^{-2} \sum_{n=m+1}^N \text{Var}(X_n) \quad N \rightarrow \infty \text{ we have}$$

$$P(\max_{n \geq m} |S_m - S_n| > \varepsilon) \leq \varepsilon^{-2} \sum_{n=m+1}^{\infty} \text{Var}(X_n) \xrightarrow{m \rightarrow \infty} 0$$

$\Rightarrow \{S_n(\omega)\}$ is a Cauchy sequence $\Rightarrow \lim_{n \rightarrow \infty} S_n(\omega)$ exists

THM 4. (Kolmogorov's three-series theorem)

Let $X_1, X_2, \dots$ i.i.d. $A > 0$ and $Y_i = X_i \mathbb{1}_{\{|X_i| \leq A\}}$.

$$\sum_{n=1}^{\infty} X_n \text{ converges a.s.} \Leftrightarrow \begin{cases} \text{(i)} & \sum_{n=1}^{\infty} P(|X_n| > A) < \infty \\ \text{(ii)} & \sum_{n=1}^{\infty} E Y_n \text{ converges} \\ \text{(iii)} & \sum_{n=1}^{\infty} \text{Var}(Y_n) < \infty \end{cases}$$

proof of sufficiency: (iii) $\xRightarrow{\text{THM 3}} \sum_{n=1}^{\infty} (Y_n - EY_n)$ converges a.s.

(ii) $\xRightarrow{\text{}} \sum_{n=1}^{\infty} Y_n$ converges a.s. (*)

(i) $\xRightarrow{\text{Borel-Cantelli Lemma}} \sum_{n=1}^{\infty} P(|X_n| > A) = \sum_{n=1}^{\infty} P(X_n \neq Y_n) < \infty \Rightarrow P(X_n \neq Y_n \text{ i.o.}) = 0$

$\Rightarrow$ have finite terms of X_n s.t. $X_n \neq Y_n \xRightarrow{(*)} \sum_{n=1}^{\infty} X_n$ converges a.s.

Lemma 2 (Kronecker's lemma)

If $a_n \uparrow \infty$ and $\sum_{n=1}^{\infty} X_n / a_n$ converges then $a_n^{-1} \sum_{m=1}^n X_m \rightarrow 0$

EXAMPLE:

THM 5 (The strong law of large numbers) *strong condition*

Let $X_1, X_2, \dots$ be i.i.d random variables with $E|X_i| < \infty$ Let $EX_i = \mu$

And $S_n = \sum_{i=1}^n X_i$ Then $\frac{S_n}{n} \rightarrow \mu$ a.s. $n \rightarrow \infty$

proof: $Y_k = X_k \cdot 1_{\{|X_k| \leq k\}}$ $T_n = Y_1 + \dots + Y_n$ then ① $\frac{T_n}{n} \rightarrow \mu$ a.s. *mean*

$\frac{S_n}{n} \rightarrow \mu$ a.s. Let $Z_k = Y_k - EY_k$ $\sum_{k=1}^{\infty} \frac{\text{Var}(Z_k)}{k^2} = \sum_{k=1}^{\infty} \frac{\text{Var}(Y_k)}{k^2} \leq \sum_{k=1}^{\infty} \frac{EY_k^2}{k^2} < \infty$ ②

$\rightarrow$ 均值为0 随机数有限
20 r.v. 收敛 a.s.

THM $\xRightarrow{EZ_k=0}$ $\sum_{k=1}^{\infty} \frac{Z_k}{k}$ converges a.s. $\xRightarrow{\text{Lemma 2}}$ $n^{-1} \sum_{k=1}^n (Y_k - EY_k) \rightarrow 0$

$\Rightarrow \frac{T_n}{n} - \frac{1}{n} \sum_{k=1}^n EY_k \rightarrow 0$ a.s. $\xrightarrow{EY_k \rightarrow EX_k = \mu, k \rightarrow \infty} \frac{T_n}{n} \rightarrow \mu$ a.s. ③

控制收敛定理 DCT

Compared with THM 2.4.1 weak condition

Let $X_1, X_2, \dots$ be pairwise i.i.d random variables with $E|X_i| < \infty$
 $EX_i = \mu$ $S_n = X_1 + \dots + X_n$ then $\frac{S_n}{n} \rightarrow \mu$ a.s. as $n \rightarrow \infty$

§ 2.5.1 Rates of Convergence

THM 6: $X_1, X_2, \dots$ i.i.d $EX_i = 0$ and $EX_i^2 = \sigma^2 < \infty$ $S_n = X_1 + \dots + X_n$

If $\varepsilon > 0$ then $S_n / n^{\frac{1}{2}(\log n)^{\frac{1}{2}+\varepsilon}} \rightarrow 0$ a.s.

proof $a_n = n^{\frac{1}{2}(\log n)^{\frac{1}{2}+\varepsilon}}$ $a_n > 0$ $\sum_{n=1}^{\infty} \text{Var}\left(\frac{X_n}{a_n}\right) = \sigma^2 \left(\frac{1}{a_1^2} + \frac{1}{n(\log n)^{1+2\varepsilon}}\right) < \infty$

THM 3 $\sum_{n=1}^{\infty} \frac{X_n}{a_n}$ converges a.s. $\xrightarrow{\text{Lemma 2}} \frac{S_n}{n^{\frac{1}{2}(\log n)^{\frac{1}{2}+\varepsilon}}} \rightarrow 0$ a.s.
 (4)

THM 7 $X_1, X_2, \dots$ i.i.d $EX_i = 0$ and $E|X_i|^p < \infty$ $1 < p < 2$
 then $S_n / n^{\frac{1}{p}} \rightarrow 0$ a.s.

proof: ① $Y_k = X_k \mathbb{1}_{\{|X_k| \leq k^{\frac{1}{p}}\}} \Rightarrow \frac{T_n}{n^{\frac{1}{p}}} \rightarrow 0$

$Z_n = Y_n - EY_n$ (central).

② To prove $\sum_{n=1}^{\infty} \frac{E(Y_n^2)}{n^{2/p}} = \sum_{n=1}^{\infty} \frac{\int_0^{\infty} 2y P(|Y_n| > y) dy}{n^{2/p}} \leq \sum_{n=2}^{\infty} \frac{\int_0^{\infty} 2y P(|X_1| > y) dy}{n^{2/p}}$
 $|Y_n| \leq |X_1|$
 $1 < p < 2$ $E|X_1|^p < \infty$

$$\textcircled{3} \sum_{m=1}^{\infty} Z_m / m^{\frac{1}{p}} \xrightarrow{\text{Lemma 2}} \frac{1}{n^{\frac{1}{p}}} \sum_{m=1}^n Z_m = \frac{1}{n^{\frac{1}{p}}} \sum_{m=1}^n (Y_m - E Y_m) \rightarrow 0 \text{ a.s.}$$

$$\textcircled{4} \frac{I_n}{n^{\frac{1}{p}}} = \frac{1}{n^{\frac{1}{p}}} \sum_{m=1}^n (Y_m - E Y_m) + \frac{1}{n^{\frac{1}{p}}} \sum_{m=1}^n E Y_m \rightarrow 0 \text{ a.s.}$$

$\rightarrow 0 \text{ a.s.} \quad ? \rightarrow 0 \text{ a.s.}$

$$\begin{aligned} \textcircled{5} E X_m = 0 &\Rightarrow E Y_m = -E[X_i; |X_i| > m^{\frac{1}{p}}] \leq E[|X_i|; |X_i| > m^{\frac{1}{p}}] \\ &= m^{\frac{1}{p}} E[|X_i| / m^{\frac{1}{p}}; |X_i| > m^{\frac{1}{p}}] \leq m^{\frac{1}{p}} E[(|X_i| / m^{\frac{1}{p}})^p; |X_i| > m^{\frac{1}{p}}] \\ &\leq m^{\frac{1}{p}+1} E[|X_i|^p; |X_i| \geq m^{\frac{1}{p}}] \end{aligned}$$

$\frac{1}{n^{\frac{1}{p}}} \sum_{m=1}^n E Y_m \rightarrow 0 \quad \textcircled{11}$

Infinite Mean:

THM 8 Let $X_1, X_2, \dots$ i.i.d with $E|X_i| = \infty$
 $S_n = X_1 + \dots + X_n$ Let a_n be a sequence of positive number
 with $\frac{a_n}{n} \uparrow$. Then $\limsup_{n \rightarrow \infty} |S_n|/a_n = 0$ or ∞ as $\sum_n P(|X_i| \geq a_n) < \infty$ or ∞

proof $\frac{a_n}{n} \uparrow \quad a_{kn} \geq k a_n \quad \forall k \quad \sum_n P(|X_i| \geq k a_n) = \infty$

$\xrightarrow{\text{B-C Lemma 7}} \{ |X_n| \geq k a_n \}$ happens infinite times (i.o.)
 X_i i.i.d.

$$\Rightarrow \limsup |X_n|/a_n = \infty \text{ a.s.} \quad \xrightarrow{|X_n| \leq |S_n| + |S_{n-1}| \leq 2 \max(|S_n|, |S_{n-1}|)} \limsup |S_n|/a_n = \infty \text{ a.s.}$$

§ 2.6 ignored

§ 2.7 large deviations

Q: $X_1, X_2, \dots$ i.i.d. $S_n = \sum_{i=1}^n X_i$ $\mu = EX_i$ when $a > \mu$

the rate at which $P(S_n > na) \rightarrow 0$

A: To prove that $\gamma(a) = \lim_{n \rightarrow \infty} \frac{1}{n} \log P(S_n \geq na)$ exists

Let $\tau_n = P(S_n \geq na)$ we have $\tau_{n+m} = P(S_{n+m} \geq (n+m)a)$
 $\geq P(S_n \geq na, S_{n+m} - S_n \geq na) = P(S_n \geq na, S_{n+m} - S_n \geq na) = \tau_n \cdot \tau_m$

$$\gamma_n = \log \tau_n \Rightarrow \gamma_{n+m} \geq \gamma_n + \gamma_m$$

Lemma 1: If $\gamma_{n+m} \geq \gamma_n + \gamma_m$ then $n \rightarrow \infty$ $\gamma_n/n \rightarrow \sup_m \gamma_m/m$

This Lemma 1 shows that $\lim_{n \rightarrow \infty} \frac{\log P(S_n \geq na)}{n} = \gamma(a)$ exists ≤ 0

According to the definition of limit $\rightarrow (\epsilon-N)$, we have $P(S_n \geq na) \leq e^{n\gamma(a)}$

However when $\gamma(a) = 0$ is not useful

We suppose that H1: $\varphi(\theta) = E \exp(\theta X_i) < \infty$ for some $\theta > 0$

Markov's inequality shows that:

$$P(S_n \geq na) = P(e^{\theta S_n} \geq e^{\theta na}) \leq \frac{E[e^{\theta S_n}]}{e^{\theta na}} \stackrel{X_i \text{ i.i.d}}{=} \frac{[E[e^{\theta X}]]^n}{e^{\theta na}}$$

$$\underbrace{\frac{k(\theta) = \log \varphi(\theta)}{\text{累积量函数}}}_{\text{累积量函数}} \rightarrow P(S_n \geq na) \leq \exp\{-n(a\theta - k(\theta))\}$$

Lemma 2: If $a > \mu$, $\theta > 0$ is small, then $a\theta - k(\theta) > 0$

proof: Consider that $k(0) = \log[\varphi(0)] = 0$ we need to prove

(i) k is continuous at 0 $\varphi(\theta) = \int e^{\theta x} dF \xrightarrow{DCT} \int 1 dF = 1$

(ii) differentiable on $(0, \theta_0)$ $\varphi'(\theta) = \lim_{h \rightarrow 0} \int \left(\frac{e^{h(x-1)}}{h}\right) e^{\theta x} dF \xrightarrow{DCT} \int \lim_{h \rightarrow 0} \dots dF$

(iii) $k'(\theta) \rightarrow \mu$ as $\theta \rightarrow 0$ $\lim_{\theta \rightarrow 0} k'(\theta) = \frac{\lim_{\theta \rightarrow 0} \int x e^{\theta x} dF(x)}{\lim_{\theta \rightarrow 0} \int e^{\theta x} dF(x)} \xrightarrow{DCT} \frac{\mu = E[X]}{1} = \mu$

在 0 附近 (右邻) $k(\theta)$ 变化率不大于 μ ($< \mu$) 积分为正。

So next step is to find the maximum of $a\theta - k(\theta)$
then an upper bound on $P(S_n \geq na)$

Let $F_\theta(x) = \frac{1}{\varphi(\theta)} \int_{-\infty}^x e^{\theta y} dF(y)$ then F_θ is a distribution function

$$E[F_\theta(x)] = \int x dF_\theta(x) = \frac{1}{\varphi(\theta)} \int_{-\infty}^{+\infty} x e^{\theta x} dF(x) = \frac{\varphi'(\theta)}{\varphi(\theta)}$$

知母函数 $M_X(t) = E[e^{tx}]$ 则 $M_X^R(t) = E[X^R e^{tx}]$

repeat the proof of Lemma 2

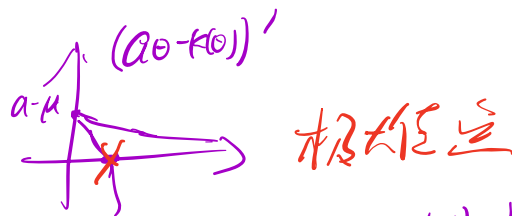
$$\psi''(\theta) = \int_{-\infty}^{\infty} x^2 e^{\theta x} dF(x)$$

$$\frac{d}{d\theta} \left(\frac{\psi'(\theta)}{\psi(\theta)} \right) = \int x^2 dF_\theta(x) - \left(\int x dF_\theta(x) \right)^2 \geq 0$$

$$\frac{\psi'(\theta)}{\psi(\theta)} \nearrow \frac{d^2}{d\theta^2} (a\theta - k(\theta)) = -\text{Var}(F_\theta) \leq 0$$

(H1) 且 $a\theta - k(\theta)$

(consider $a = \frac{\psi'(\theta)}{\psi(\theta)} = k'(\theta)$)



极大值点

EXAMPLE: $X \sim N(0, 1)$ $\mu = 0$ (完美情况, 解总存在)

$$\psi(\theta) = \exp\left(\frac{\theta^2}{2}\right) \quad k(\theta) = \log \psi(\theta) = \frac{\theta^2}{2} \quad \frac{\psi'(\theta)}{\psi(\theta)} = \theta$$

是单峰的

$$a\theta - k(\theta) = a\theta - \frac{\theta^2}{2} \text{ 二次函数, 开口向下凹性.}$$

$\theta_a = a$ 为极大值点

$F_\theta \sim N(\theta, 1) \rightarrow$ 倾斜分布.
 let $\theta = \theta_a = a$ the mean of $F_\theta = a$

" $a = \frac{\psi'(\theta)}{\psi(\theta)}$ 的意义": 寻找参数 θ , 使给分布经指数倾斜变换后
新分布的均值等于大偏差问题的阈值 a

此例子 θ_a 总存在, 还有一些例子中 a 在某些情况下 θ_a 才存在

THM1 Suppose in addition to (H1) and (H2) that

there is a $\theta_a \in (0, \theta_+)$ so that $a = \frac{\psi'(\theta_a)}{\psi(\theta_a)}$ Then $n \rightarrow +\infty$,

$$n^{-1} \log P(S_n \geq na) \rightarrow -a\theta_a + \log(\varphi(\theta_a))$$

proof ① $\limsup_{n \rightarrow \infty} n^{-1} \log P(S_n \geq na) \leq -a\theta_a + \log(\varphi(\theta_a)) \checkmark$

② $\liminf_{n \rightarrow \infty} n^{-1} \log P(S_n \geq na) \geq -a\theta_a + \log(\varphi(\theta_a))$

取 $\theta \in (\theta_a, \theta_1)$ 定义 new prob distr F_λ $\frac{dF}{dF_\lambda} = e^{-\lambda x \varphi(\lambda)}$

is the Radon-Nikodym derivative of associated measure
 $\lambda \in (\theta_a, \theta_1)$ then $M_\lambda = \frac{\varphi(\lambda)}{\varphi(1)} > a$

If we let F_λ^n, F^n denote S_n^λ, S_n , we have $\frac{dF^n}{dF_\lambda^n} = e^{-\lambda x \varphi(\lambda)^n}$

($n=1$ 时 P 为 δ_a) $\boxed{\forall v > a}$

Then $P(S_n \geq na) = \int_{x \geq na} dF^n(x) \geq \int_{na}^{\infty} e^{-\lambda x [\varphi(\lambda)]^n} dF_\lambda^n(x)$
 $\geq [\varphi(\lambda)]^n e^{-\lambda n v} \int_{na}^{nv} dF_\lambda^n(x)$

$$\log P(S_n \geq na) \geq n \log(\varphi(\lambda)) - \lambda n v + \log [F_\lambda^n(nv) - F_\lambda^n(na)]$$

$$\liminf_{n \rightarrow \infty} \log P(S_n \geq na) \geq n \log(\varphi(\lambda)) - \lambda n v + \underbrace{\liminf_{n \rightarrow \infty} [F_\lambda^n(nv) - F_\lambda^n(na)]}_{\text{term}}$$

$$\lim_{n \rightarrow \infty} P(na \leq S_n^\lambda \leq nv) \xrightarrow{v > M_\lambda} \lim_{n \rightarrow \infty} [F_\lambda^n(nv) - F_\lambda^n(na)] = 1$$

Finally: $\liminf_{n \rightarrow \infty} n^{-1} \log P(S_n \geq na) \geq \log(\varphi(\theta_a)) - \theta_a a$

□